



A.2 Forces and momentum

Space, time and motion · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Newton's three laws govern how forces change motion: an object's velocity is unchanged unless acted on by a net force (1st law); net force equals rate of change of momentum, $F = \Delta p / \Delta t = ma$ for constant mass (2nd law); forces come in equal, opposite, colinear action-reaction pairs on different bodies (3rd law). Linear momentum $p = mv$ is conserved in any isolated system, which underlies collision analysis: in elastic collisions both momentum and kinetic energy are conserved, while in inelastic collisions only momentum is conserved. Impulse, $J = F\Delta t = \Delta p$, equals the area under a force-time graph.

Worked example: follow the reasoning

A 0.50 kg ball moving at $6.0 \text{ m}\cdot\text{s}^{-1}$ collides head-on with a stationary 1.5 kg ball. After collision the 0.50 kg ball moves at $-2.0 \text{ m}\cdot\text{s}^{-1}$ (rebounds). Find the velocity of the 1.5 kg ball.
Conservation of momentum: $0.50(6.0) + 0 = 0.50(-2.0) + 1.5v$
 $3.0 = -1.0 + 1.5v \rightarrow v = 2.67 \text{ m}\cdot\text{s}^{-1}$.
Answer: $2.7 \text{ m}\cdot\text{s}^{-1}$ in the original direction.

Quick check · multiple choice

A constant force of 10 N acts on a 2.0 kg mass for 4.0 s, starting from rest. What is the final momentum?
A $5 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
B $20 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
C $40 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
D $80 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A 1200 kg car travelling at $15 \text{ m}\cdot\text{s}^{-1}$ collides with a stationary 900 kg car; they lock together on impact. [6]
(a) Calculate the common velocity immediately after the collision.
(b) Calculate the kinetic energy lost in the collision.
(c) State, with a reason, whether this collision is elastic or inelastic.

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) Momentum conservation: $1200(15) + 0 = (1200+900)v \rightarrow v = 18000/2100 = 8.57 \text{ m}\cdot\text{s}^{-1}$.
- (b) KE before = $\frac{1}{2}(1200)(15^2) = 135000 \text{ J}$. KE after = $\frac{1}{2}(2100)(8.57^2) = 77143 \text{ J}$. Loss = $135000 - 77143 \approx 57900 \text{ J} \approx 5.79 \times 10^4 \text{ J}$.
- (c) Inelastic, because kinetic energy is not conserved (some converts to heat/sound/deformation) even though momentum is conserved.

Quick check answer

C. Impulse $J = F\Delta t = 10 \times 4.0 = 40 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} = \Delta p$; starting from rest, final $p = 40 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.